

Group Project 1: Statistical Analysis of a Currency (Japan)

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ECO 442

Part 1: FRED Graph

The following figure shows the daily nominal exchange rate for Japan against the U.S. dollar over the past 10 years. Denoted as:

$$E_{\text{yen}/\$}$$



Figure 1: Japanese Yen per U.S. Dollar, daily nominal exchange rate.

Part 2: Hotel

Japan's capital city is Tokyo. The hotel we chose is the Trunk Cat Street. The cost for one night is:

$$P_{\text{yen}} = \text{¥}669,944.$$

Part 3: Hotel Price

To convert from yen into dollars:

$$P_{\$} = \frac{P_{\text{yen}}}{E_{\text{yen}/\$}}.$$

(1) **Wed Nov 26:** $E_{\text{yen}/\$} = 156.38$

$$P_{\$}^{\text{Nov } 26} = \frac{669,944}{156.38} = 4284.08, \quad \Rightarrow \quad P_{\$}^{\text{Nov } 26} \approx \$4,284.08$$

(2) **Fri Nov 28:** $E_{\text{yen}/\$} = 156.17$

$$P_{\$}^{\text{Nov } 28} = \frac{669,944}{156.17} = 4289.84, \quad \Rightarrow \quad P_{\$}^{\text{Nov } 28} \approx \$4,289.84$$

Part 4: Appreciation/depreciation

The exchange rate fell from 156.38 to 156.17 yen per dollar. Since $E_{\text{yen}/\$}$ is yen per \$1, a decrease in $E_{\text{yen}/\$}$ means the yen appreciated.

Holding the hotel price fixed at ¥669,944, the U.S. dollar price increased from approximately \$4,284.08 to \$4,289.84. The change in the dollar price was:

$$\$4,289.84 - \$4,284.08 = \$5.76.$$

The yen's appreciation made the hotel \$5.76 more expensive for a U.S. tourist on Nov 28 compared to Nov 26.